

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		Attempt at a tangent at $t = 1\text{ s}$ (1) Correct values taken from graph (1) $F = 0.035\text{ [N]}$ [tolerance ± 0.01] (1)	1 1	1		3	3	
	(b)		Graph horizontal / momentum does not change / constant before impact (1) Indicates no resultant / external / net forces (1) Conservation of momentum applies (1) [Gain in momentum of B is equal to the loss in A]	1	1 1		3		
	(c)	(i)	At zero between 0 s and 2.8 s - accept no line by implication (1) Straight line increase of 0.08 N s between 2.8 s and 3.0 s (1) ecf Horizontal at 0.08 N s from 3.0 s to 4.0 s (1) ecf		3		3	3	
		(ii)	$p_B = 0.08$ ecf (1) $v = \frac{0.08}{0.16} = 0.5\text{ [ms}^{-1}\text{]}$ (1)		2		2	2	
			Question 6 total	3	8	0	11	8	0